

PARENT, QC, Canada

WMO: 717260

Lat: 47.922N

Lon: 74.624W

Elev: 445

StdP: 96.10

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -34.1	(c) -30.9	(d) -38.5	(e) 0.1	(f) -33.8	(g) -35.2	(h) 0.1	(i) -30.5	(j) 6.8	(k) -12.9	(l) 6.1	(m) -12.7	(n) 0.7	(o) 290	(p) 0.582

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	27.8	19.1	26.0	18.5	24.3	17.6	21.3	25.3	20.1	23.8	19.1	22.6	2.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.0	15.5	23.1	18.9	14.5	22.0	17.8	13.5	21.0	64.1	25.6	60.0	23.9	56.3	22.7	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 5.7	(b) 5.0	(c) 4.4	(e) -38.4	(f) 30.9	(g) 3.5	(h) 1.4	(i) -41.0	(j) 31.9	(k) -43.0	(l) 32.7	(m) -45.0	(n) 33.5	(o) -47.6	(p) 34.6		
(a) 5.7	(b) 5.0	(c) 4.4	(e) -38.5	(f) 22.9	(g) 3.2	(h) 1.4	(i) -40.8	(j) 23.9	(k) -42.7	(l) 24.7	(m) -44.5	(n) 25.5	(o) -46.9	(p) 26.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.8	-16.1	-14.3	-7.8	0.3	9.2	14.5	17.0	15.6	11.4	4.8	-2.9	-10.9
	DBStd	12.86	8.20	7.40	7.47	5.41	4.84	4.24	3.21	3.33	4.41	4.74	5.74	7.20
	HDD10.0	3656	808	679	552	291	74	8	0	2	35	173	387	648
	HDD18.3	6077	1066	912	810	539	284	128	65	96	213	419	637	906
	CDD10.0	678	0	0	0	2	50	144	218	176	76	13	0	0
	CDD18.3	57	0	0	0	0	3	14	24	12	4	0	0	0
	CDH23.3	575	0	0	0	1	58	173	208	106	28	2	0	0
	CDH26.7	102	0	0	0	0	12	38	36	13	4	0	0	0
Wind	WSAvg	1.8	1.9	1.9	2.0	2.1	1.9	1.5	1.4	1.3	1.4	1.9	2.0	1.8
Precipitation	PrecAvg	1035	86	53	58	71	72	94	114	101	102	91	93	103
	PrecMax	1559	279	104	138	109	129	169	225	183	207	149	261	379
	PrecMin	805	23	21	28	42	23	33	25	24	39	33	21	38
	PrecStd	176	54	24	22	20	26	33	44	39	45	32	45	73
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.2	5.5	13.1	19.4	28.2	30.1	29.8	28.4	26.4	21.7	12.9	5.0
		MCWB	6.0	4.0	6.8	11.4	16.3	20.1	20.6	20.2	19.7	15.8	11.0	4.3
	2%	DB	1.7	1.7	7.8	14.8	24.7	27.6	27.5	26.1	23.5	17.7	9.2	2.2
		MCWB	1.4	0.2	3.3	8.0	14.6	18.5	19.7	19.4	18.1	12.9	7.5	1.6
	5%	DB	-1.2	-0.3	5.0	11.7	21.9	25.3	25.6	24.3	21.2	14.9	6.7	0.6
		MCWB	-1.8	-1.3	1.5	6.4	13.0	17.2	19.1	18.4	16.9	11.8	5.4	0.1
	10%	DB	-3.7	-2.4	2.8	8.9	18.9	22.9	23.9	22.4	19.2	12.4	4.4	-0.8
		MCWB	-4.4	-3.4	0.6	4.4	11.9	16.2	18.1	17.5	15.7	10.0	3.3	-1.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.1	3.8	8.7	12.8	19.0	22.4	23.6	22.1	20.7	16.6	11.2	4.4
		MCDB	6.3	4.3	11.7	18.3	24.2	27.1	26.5	26.4	24.5	20.2	12.2	4.7
	2%	WB	1.5	0.1	4.0	9.4	16.6	20.4	21.6	20.5	19.3	14.1	8.0	1.8
		MCDB	1.8	1.0	6.2	13.1	21.6	24.6	25.4	24.4	22.2	17.0	8.8	2.2
	5%	WB	-1.7	-1.5	2.2	7.1	14.8	19.0	20.3	19.4	17.8	12.2	5.5	0.3
		MCDB	-1.2	-0.6	4.5	10.5	19.5	22.9	23.8	22.8	20.6	14.5	6.4	0.6
	10%	WB	-4.4	-3.5	0.4	5.0	12.9	17.6	19.1	18.2	16.1	10.1	3.4	-1.3
		MCDB	-3.8	-2.6	2.4	8.4	17.5	21.4	22.8	21.2	18.7	12.0	4.3	-0.8
Mean Daily Temperature Range	5% DB	MDBR	11.8	13.2	12.8	11.4	13.0	12.3	11.2	10.9	10.7	8.2	7.2	9.0
		MCDBR	10.3	11.3	14.0	16.2	18.5	15.6	13.8	13.8	13.3	12.7	10.0	7.2
		MCWBR	10.3	10.3	10.2	10.2	9.3	7.6	7.4	7.5	8.2	8.7	8.5	7.1
	5% WB	MCDBR	10.5	10.7	11.8	14.0	14.4	12.6	11.4	11.5	11.2	11.3	9.1	6.7
		MCWBR	10.6	10.0	9.1	10.0	8.4	7.3	7.0	7.1	7.7	8.5	8.4	6.7
Clear-Sky Solar Irradiance	taub	0.269	0.279	0.298	0.377	0.345	0.360	0.378	0.359	0.353	0.319	0.292	0.261	
	taud	2.229	2.219	2.230	2.196	2.449	2.428	2.389	2.440	2.446	2.535	2.477	2.320	
	Ebn at Noon	801	881	919	872	918	901	878	883	854	837	786	772	
	Edh at Noon	77	98	115	133	108	111	114	104	94	73	62	63	
All-Sky Solar Radiation	RadAvg	1.28	2.26	3.65	4.59	5.10	5.53	5.19	4.47	3.48	1.98	1.07	0.89	
	RadStd	0.19	0.34	0.35	0.35	0.46	0.40	0.55	0.29	0.35	0.25	0.15	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(a) Station Only	(b) Regional (5 neighbors)	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(a) Station Only	(b) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page